

Project 1: Create and Access an EC2 Instance Using SSH

Introduction

Cloud computing provides on-demand access to computing resources such as servers, storage, networking, and software over the internet. Amazon Web Services (AWS) is one of the leading cloud service providers. One of its core services is Amazon EC2 (Elastic Compute Cloud), which allows users to create and manage virtual servers in the cloud.

This project focuses on launching an EC2 instance and securely accessing it using SSH.

Objective

The objective of this project is to:

- ✧ Launch a virtual server using Amazon EC2
- ✧ Configure secure access using a key pair
- ✧ Configure network security using Security Groups
- ✧ Connect to the EC2 instance using SSH
- ✧ Demonstrate complete instance lifecycle management

Tools and Services Used

- ✧ Amazon Web Services (AWS) Console
- ✧ Amazon EC2
- ✧ Amazon Linux 2023
- ✧ Security Groups
- ✧ RSA Key Pair (.pem format)
- ✧ Windows Command Prompt (SSH Client)

Project Architecture

The architecture consists of:

- ✧ A single EC2 instance deployed in the ap-southeast-2 (Sydney) region
- ✧ A Security Group allowing SSH access (Port 22) from the user's IP address
- ✧ An RSA key pair used for authentication
- ✧ Secure SSH connection from local system to EC2 instance

Implementation Steps

Step 1: AWS Login and Region Selection

Logged into the AWS Management Console and selected the ap-southeast-2 (Sydney) region for deployment.

Step 2: Launch EC2 Instance

- ✧ Navigated to EC2 Dashboard
- ✧ Clicked "Launch Instance"
- ✧ Named the instance: Internship-EC2-SSH-Instance
- ✧ Selected Amazon Linux 2023 AMI
- ✧ Selected instance type t3.micro
- ✧ Created a new RSA key pair in .pem format

Step 3: Configure Security Group

Created a new security group
Allowed inbound SSH traffic (Port 22)
Set source as "My IP" for secure access

Step 4: Instance Launch and Verification

- ✧ Launched the instance
- ✧ Verified instance state as "Running"
- ✧ Confirmed status checks (2/2 passed)

Step 5: SSH Connection

- ✧ Updated .pem file permissions using `icacs` command
- ✧ Connected to EC2 using SSH:

```
ssh -i Internship-EC2-KeyPair.pem ec2-user@<Public-IP>
```

- ✧ Verified successful login into Amazon Linux shell

Step 6: Instance Lifecycle Management

- ✧ Stopped the instance to avoid billing
- ✧ Terminated the instance after project completion

Results

The EC2 instance was successfully launched and securely accessed via SSH. The project demonstrated proper configuration of key pairs, security groups, and secure authentication. The instance lifecycle was properly managed to ensure cost control.

Challenges Faced

- ✧ Configuring correct file permissions for the .pem file on Windows
- ✧ Understanding SSH host authenticity prompt
- ✧ Managing instance state and public IP changes

Key Learnings

- ✧ Understanding the working of Amazon EC2
- ✧ Importance of Security Groups in network protection
- ✧ Role of key pairs in secure authentication
- ✧ SSH connectivity process
- ✧ Resource lifecycle management in cloud computing
- ✧ Cost awareness in AWS

Conclusion

This project successfully demonstrated the process of launching and accessing a virtual server using Amazon EC2. It provided hands-on experience in cloud infrastructure provisioning, security configuration, and remote server management. The project builds foundational knowledge required for advanced cloud computing and infrastructure automation tasks.